

Correction of the 20,000-bin binning method (MWI 1997 / 2004, BEL 2000)

1. The original problem

PIP's code (`get_refy_quantiles()`, in `lineup_estimate.R`) builds the 20,000 bins like this:

```
Qx <- fquantile(x$welfare, w = x$weight, probs = probs, names = FALSE)
data.table(welfare = Qx, weight = xpop / nobs, reporting_level = rl)
```

- It takes 20,000 **quantile-function points** (`fquantile`) and uses them as if they were the mean of each bin.
- It assigns the same weight `xpop / nobs` to each bin.

This is **not the same** as collapsing the distribution into bins while preserving the mean. The original mean is:

$$\mu = \frac{\sum_i w_i y_i}{\sum_i w_i}$$

but the mean produced by the quantile method is approximately:

$$\hat{\mu} = \frac{1}{n} \sum_j Q(p_j)$$

which is a numerical approximation (a Riemann-sum type) to the integral of the quantile function. With heavy tails, censoring, or extreme values, that approximation can deviate noticeably from the true mean.

2. The solution: Lorenz-curve-based bins

`get_refy_quantiles_mean()` (in `build_20k_bins_2004.R` / `build_20k_bins_1997.R`) builds the bins differently: it first defines bins of **equal population weight** and then computes the **actual average welfare within each bin**, using the generalized Lorenz curve:

1. `p_bounds` — 20,001 equally-spaced probability cutpoints (from 0 to 1).
2. `p_cum` — cumulative population share: `cumsum(weight) / xpop`.
3. `cum_inc` — cumulative **total** welfare (generalized Lorenz curve): `cumsum(weight * welfare)`.
4. `approx(p_cum, cum_inc, xout = p_bounds)` — linearly interpolates the exact cumulative welfare at each of the 20,000 cutpoints. Linear interpolation **splits** observations (households) that straddle two bins, the same way `lorenz_table()/new_bins` does in the original pipeline.
5. `bin_inc <- diff(interpolated_inc)` — total welfare inside each bin.
6. `bin_welfare <- bin_inc / (xpop / nobs)` — the bin's mean = its total welfare / its population.

Why it preserves the mean: each bin retains its total weight and total welfare before computing the average, so the weighted mean of the 20,000 bins is mathematically identical to the weighted mean of the original microdata (not an approximation).

Empirical validation

Dataset	Weighted mean of microdata	Weighted mean of 20,000 bins
MWI 2004 (<code>mwi_2004_reportingyear_processed.dta</code>)	2.844316	2.844316
MWI 1997 (<code>mwi_1997_reportingyear_processed.dta</code>)	5.724018	5.724018

They match exactly, not just approximately.

3. The prerequisite: raw data doesn't work directly

The binning assumes `welfare` is **already in the correct final unit** (2021 PPP \$/day). If the raw survey (nominal local currency) is used, the binning still preserves the mean — but it preserves the mean of a variable on the wrong scale:

- `mwi_2004_reportingyear_processed.dta` (already processed) → worked directly, reasonable mean (~2.84).
- `MWI_1997.dta` raw → gave an absurd mean (~12,808), because `welfare` was still in nominal local currency, without dividing by CPI/ICP/365 or adjusting by the lineup factor.

4. The "lineup" process: 1997 and 2004 are different

1997 is its own survey year (reporting year 1997 == survey year 1997.83). `interpolated_means.dta` has a **single row** for `year == 1997` (`relative_distance = 1.0`), meaning: **there is no interpolation or extrapolation between two surveys**. It only goes through PPP 2021 conversion and its own `mult_factor` (`predicted_mean_ppp/svy_mean` from that same row).

2004 does combine two surveys (1997.83 and 2004.23), because reporting year 2004 falls "between" both. Each survey is rescaled according to its `relative_distance` to the reporting year (2004: ≈ 0.964 for the 2004 survey, ≈ 0.036 for the 1997 one), and both are pooled into a single dataset representing reporting year 2004.

This is done by:

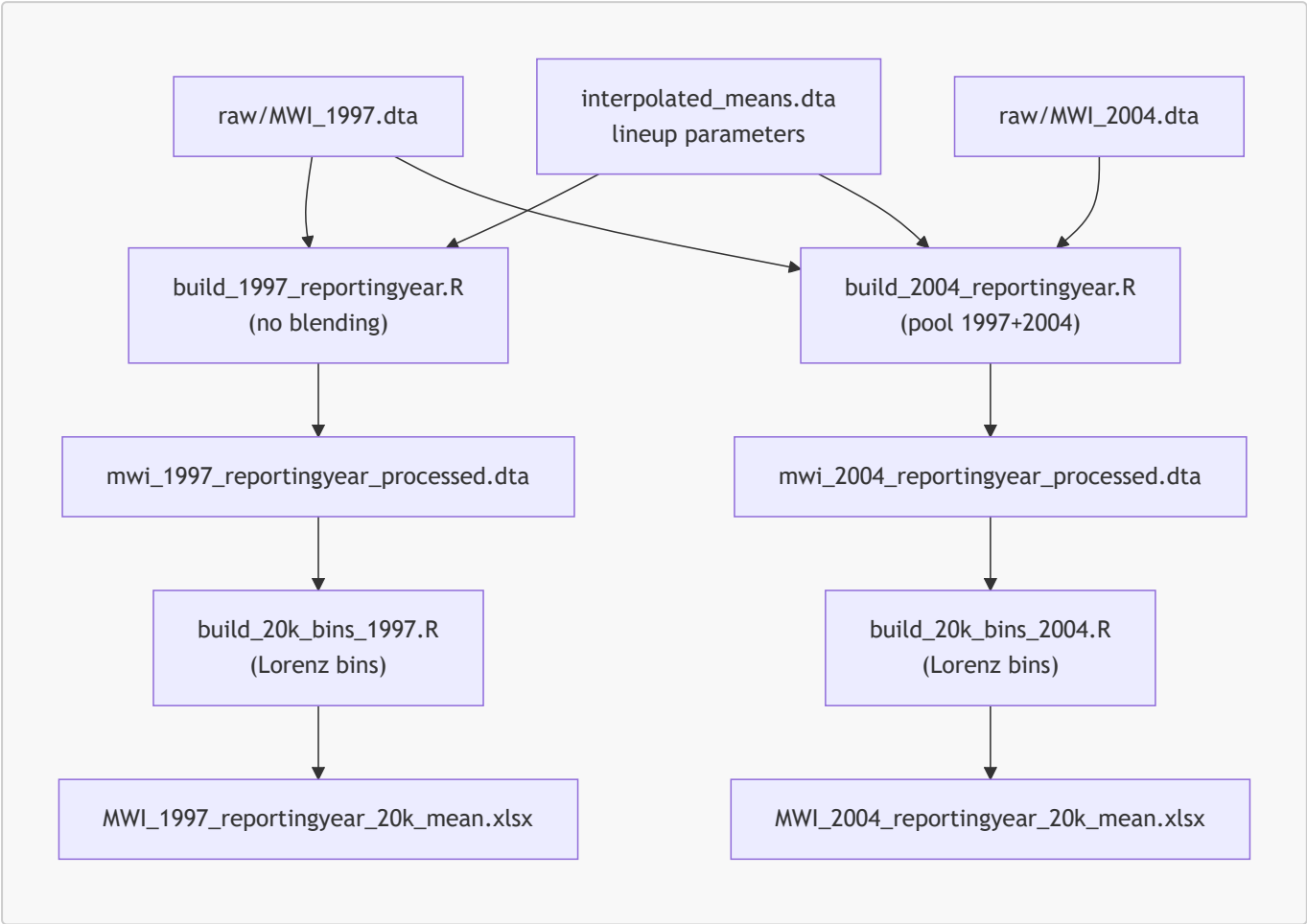
- `build_1997_reportingyear.R` → uses only `raw/MWI_1997.dta` → generates `mwi_1997_reportingyear_processed.dta`.
- `build_2004_reportingyear.R` → uses `raw/MWI_1997.dta` + `raw/MWI_2004.dta` → generates `mwi_2004_reportingyear_processed.dta`.

Steps (common to both, but with different lineup info):

1. **Lineup parameters:** from `01-input/interpolated_means.dta`, take `nac`, `nac_sy`, `relative_distance`, `predicted_mean_ppp`, `svy_mean`, `reporting_pop` for the corresponding reporting year.
2. **Conversion to PPP/day:** `welfare_ppp = welfare / (cpi2021 * icp2021 * 365)`.
3. **Lineup factor:** `mult_factor = predicted_mean_ppp / svy_mean` — rescales each survey so its mean matches the interpolated mean for the reference year.

4. **Population weight rescaling:** `weight = weight * (reporting_pop / svy_pop) * relative_distance.`
5. **Applying the factor:** `welfare = welfare_ppp * mult_factor.`
6. **Bottom coding:** `welfare = 0.28` if `welfare < 0.28.`

5. Full end-to-end pipeline



6. Process files

File	Role
<code>build_1997_reportingyear.R</code>	Processes only the 1997 survey (reporting year 1997, no blending). Generates <code>mwi_1997_reportingyear_processed.dta</code> .
<code>build_2004_reportingyear.R</code>	Combines (pools) the 1997 + 2004 surveys for reporting year 2004. Generates <code>mwi_2004_reportingyear_processed.dta</code> .
<code>build_20k_bins_1997.R</code>	Lorenz-curve-based binning method for reporting year 1997.
<code>build_20k_bins_2004.R</code>	Lorenz-curve-based binning method for reporting year 2004.
<code>01-processing/raw/MWI_1997.dta, MWI_2004.dta</code>	Raw input microdata.

File	Role
01- processing/mwi_1997_reportingyear_processed.dta	Processed data, reporting year 1997 (no blending).
01- processing/mwi_2004_reportingyear_processed.dta	Processed data, reporting year 2004 (pool of 1997+2004).
03-outputs/MWI_1997_reportingyear_20k_mean.xlsx /MWI_2004_reportingyear_20k_mean.xlsx	Final results: 20,000 bins with the mean preserved.

7. Conclusion

- The binning method (`get_refy_quantiles_mean`) is correct and preserves the mean exactly, unlike the point-quantile method (`fquantile`), which only approximates it.
- But that method **requires welfare to already be in its final unit** (\$ PPP/day, adjusted by lineup). If given the raw survey without that prior processing, the binning still "works" (it doesn't error out) but produces a mean on the wrong scale.
- 1997 and 2004 require different processing: 1997 (its own survey year) only needs PPP/day + its own `mult_factor`; 2004 additionally needs the pooling and rescaling of two surveys.
- The correct order is: **1)** process the raw survey with the lineup logic corresponding to the reporting year, **2)** apply the Lorenz-based binning on that already-processed data.

8. Special case: BEL 2000 (raw data already comes as 1,000 bins)

Unlike MWI (individual microdata, one record per household/person), `raw/BEL_2000.dta` **already comes pre-aggregated into 1,000** population bins (one row per bin, each with its average `welfare` and the same `weight`). This changes the process in two ways:

a) Its own reporting year, no blending. Just like MWI 1997, `interpolated_means.dta` has a single row for `country_code == "BEL" & year == 2000` (`relative_distance = 1.0`, `predicted_mean_ppp == svy_mean`, so `mult_factor = 1`). It only goes through PPP 2021 conversion, weight rescaling to `reporting_pop`, and bottom coding — no interpolation/extrapolation between two surveys.

b) Going from 1,000 to 20,000 bins by duplication, not microdata. Since there is no information about the distribution *within* each of the 1,000 bins (only their average), the Lorenz-curve interpolation can't be applied directly over just 1,000 points without distorting the original structure. Instead, each of the 1,000 bins is **duplicated 20 times** (same `welfare`, `weight` split into 20 equal parts), generating 20,000 rows of pseudo-microdata that preserve exactly the mean and the shape of the distribution (same Lorenz curve, just at a finer resolution). The same `get_refy_quantiles_mean()` function used for MWI is then run on that pseudo-microdata, for consistency of method.

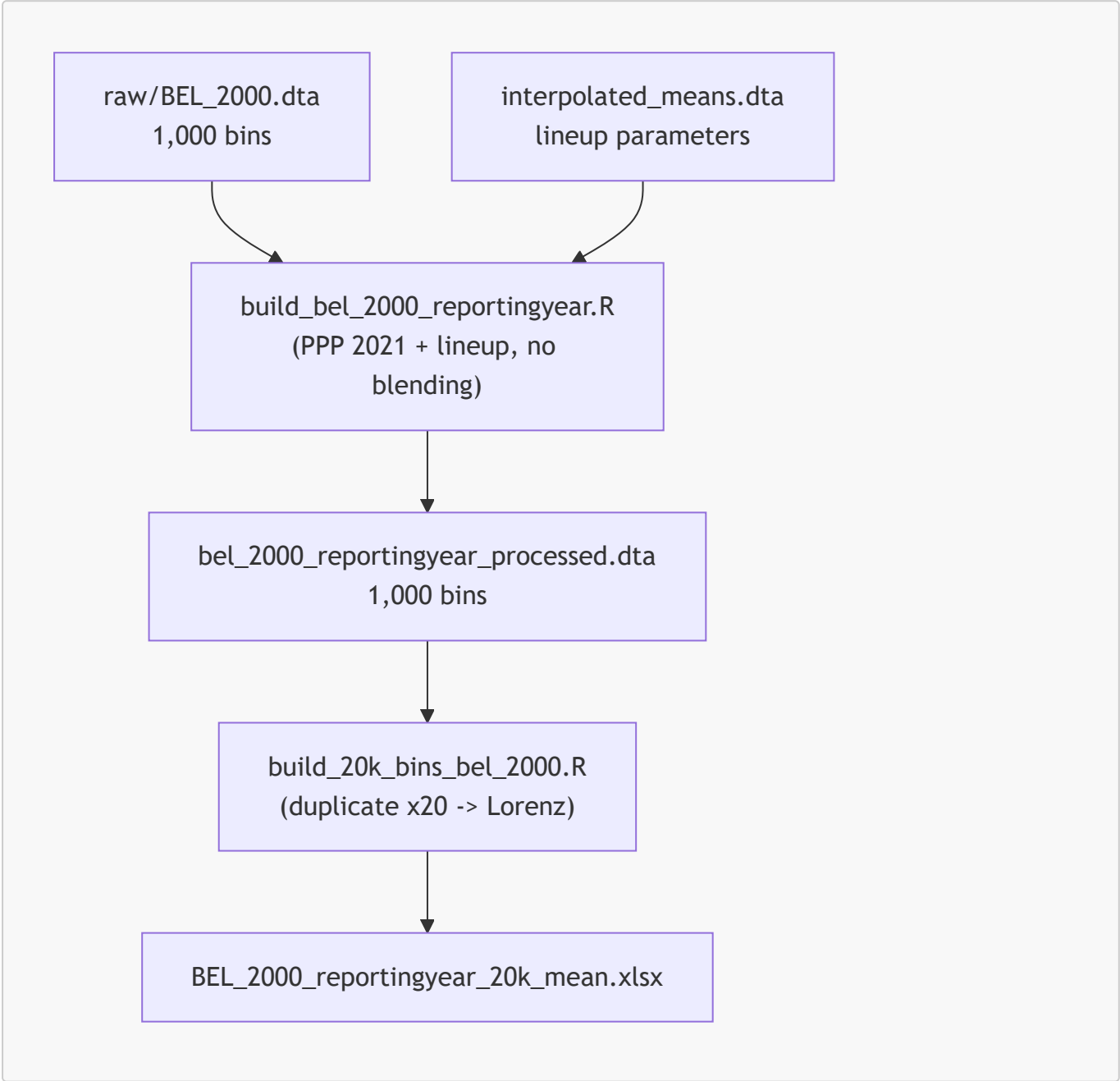
Empirical validation (BEL 2000)

Stage	Rows	Weighted mean
Processed 1,000 bins (<code>bel_2000_reportingyear_processed.dta</code>)	1,000	63.42967
Expanded pseudo-microdata (x20)	20,000	63.42967

Stage	Rows	Weighted mean
Final 20,000 bins (BEL_2000_reportingyear_20k_mean.xlsx)	20,000	63.42967

The mean is preserved exactly at every stage.

BEL 2000 pipeline



BEL 2000 process files

File	Role
build_bel_2000_reportingyear.R	PPP 2021 + lineup (no blending) on the raw 1,000 bins. Generates bel_2000_reportingyear_processed.dta.
build_20k_bins_bel_2000.R	Duplicates each bin 20x (1,000 → 20,000 rows) and applies the Lorenz-based binning

File	Role
	method.
01-processing/raw/BEL_2000.dta	Raw data: 1,000 population bins (not individual microdata).
01-processing/bel_2000_reportingyear_processed.dta	1,000 bins already in 2021 PPP/day.
03-outputs/BEL_2000_reportingyear_20k_mean.xlsx	Final result: 20,000 bins with the mean preserved.